

Roll No.

TPE-301

B. TECH. (PETROLEUM ENGINEERING) (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

THERMODYNAMICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Define the following : (CO1)

(i) First law of thermodynamics

(ii) Show that $C_p - C_v = R$ for the ideal gas.

(iii) Intensive and Extensive Properties with examples

(iv) Degree of freedom. What will be DOF for the water at its triple point ?

(v) Internal energy and Enthalpy

P. T. O.

Roll No.

TPE-302

**B. TECH. (PETROLEUM ENGINEERING)
(THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023**

APPLIED GEOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) What are the seismic waves ? Describe the different types of seismic waves. (CO1)

OR

- (b) Describe the internal structure of earth. (CO1)

2. (a) Differentiate between exogenic and endogenic processes by giving two examples of each processess. (CO1)

OR

- (b) What is weathering ? Differentiate between weathering and erosion. List and explain the factors affecting weathering. (CO1)

P. T. O.

(2)

3. (a) What is physical weathering ? Describe the processess through which physical weathering takes place. (CO1)

OR

- (b) What is chemical weathering ? Describe the processess through which chemical weathering takes place. (CO1)

4. (a) What are the processess through which river erosion take place ? (CO1)

OR

- (b) What is the theory of continental drift ? What evidences Wegner cited in support of his theory ? Why the theory was not accepted at that time ? (CO1)

5. (a) What are the characteristics of a mineral ? Describe the physical properties of mineral used for hand speciment identification of minerals. (CO2)

OR

- (b) Define rock. Explain how one type of rock is converted into another by drawing rock cycle. (CO3)

Roll No.

TPE-303

B. TECH. (PE) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

FLUID AND PARTICLE MECHANICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What is tensile, compressive and shear stress ? (CO1)

OR

(b) What is an incompressible fluid ? Explain modulus of compressibility. (CO1)

2. (a) What do you understand by viscosity ? What is its relation with the rate of shear strain ? (CO1)

OR

(b) What is shear stress and how is it related to the coefficient of viscosity ? (CO1)

3. (a) What is body force ? Give *three* examples. (CO1/CO2)

P. T. O.

(2)

OR

- (b) Derive hydrostatic law. By how much will the pressure change if the depth of fluid is increased by two times ? (CO1/CO2)
4. (a) Explain with diagram how can the pressure be measured with a manometer. (CO2)

OR

- (b) Explain what is gauge pressure, vacuum pressure and absolute pressure ? (CO2)
5. (a) Derive continuity equation for steady flow in a pipe. (CO2)

OR

- (b) Explain the difference between steady and unsteady flow. Give *one* example of each. (CO2)

Roll No.

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MID Semester Examination Oct 2023

Name of the Program: B.Tech. (PE)
Name of the Course: Geomatics Engineering

Semester: III
Paper Code: TPE 304

Time: 90 minutes

Maximum Marks: 50

Note: -

- (i) Answer all questions by choosing any one of the sub questions
- (ii) Each question carries 10 marks

Q1. (10 Marks) [CO1]

- a. Explain two methods each of ranging on smooth leveled ground and on sloping ground.

OR

- b. Define Plan and Map and how do they differ from each other. Explain a graphical scale and Representative fraction?

Q2. (10 Marks) [CO1]

- a. A base line is measured with a steel tape, it is approximately 1000m long. Calculate the correct length of the baseline at MSL when the pull at the standardization equal 15kg. The pull applied is 23kg. Cross sectional area of the tape is 0.0645 cm^2 , $E = 2.11 \times 10^6 \text{ kg/cm}^2$. Temperature at the time of measurement was 35°C and standardized at 15°C respectively. The difference of level between the two ends of the base line is 2.0 m, Radius of earth $R = 6400 \text{ km}$. Elevation of base line above MSL is 1000 m, $\alpha = 12 \times 10^{-6}/^\circ \text{C}$. Determine the correct length of line after applying the necessary corrections as per the data given in the numerical.

OR

- b. A plan of an old survey plotted to a scale of 10m to 1cm carried a note stating that, the chain (20m) used was 0.8 links (16cm) too short. It was also found that the plan had shrunk so that a line originally 10cm long was 9.77 cm. The area of a plot available on the plan was found to be 58.2 cm^2 . What is the correct area of the plan?

Q3. (10 Marks) [CO1]

- a. Explain any two of the following terms related to chain surveying with the help of neat diagram: Line ranger, Cross staff, Optical Square, Clinometer.

OR

- b. Explain difference between Prismatic compass and Surveyor's compass with a sketch.

Q4.

(10 Marks) [CO2]

- a. Enlist the temporary adjustments to be carried out for theodolite? Explain three screw head method of levelling.

OR

- b. A 20 m chain was found to be 10 cm too long after chaining a distance of 1500 m. It was found to be 18 cm too long at the end of day's work after chaining a distance of 2900 m. Find the true distance if the chain was correct before commencement of the work.

Q5.

(10 Marks) [CO2]

OR

- a. The following bearings were observed in a compass traverse over an area where local attraction was suspected. Find the amount of local attraction at different stations, the correct bearing of the line and the included angles. Also Draw the sketch on a scale AB = 100 m, BC = 100 m and CD = 50 m.

Line	FB	BB
AB	68° 15'	248° 15'
BC	148° 45'	326° 15'
CD	224° 30'	46° 00'
DE	217° 15'	38° 15'
EA	327° 45'	147° 45'

OR

- c. A traverse ABCDA is made in the form of a square taking in clockwise order. If the bearing of AB is 120° 30', find the bearing of other sides.

Roll No.

TPE-311

B. TECH. (PETROLEUM ENGINEERING)
(THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023
INTRODUCTION TO PETROLEUM OPERATIONS

Time : 1½ Hours

Maximum Marks : 50

- Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) What is petroleum ? Describe the chemical composition of crude oil.

(CO1)

OR

- (b) Describe the main operations involved in upstream, mid-stream and downstream oil and gas industry.

(CO1)

2. (a) What do you mean by petroleum exploration ? List the different methods of petroleum exploration. Explain gravity method of exploration.

(CO1)

OR

- (b) List the salient feature of Geo-Technical Order (GTO).

(CO1)

P. T. O.

(2)

3. (a) What is drilling ? List the different types of drilling methods. What are the main systems in a rotary drilling methods ? (CO2)

OR

- (b) What is directional drilling techniques ? List and describe the *four* applications of directional drilling. (CO2)
4. (a) What are the components of a hosting system ? Explain the following operations in drilling : (CO3)
- (i) Making a connection
 - (ii) Making a trip
 - (iii) Wiper trip

OR

- (b) Describe the classification of well used in oil industry. (CO3)
5. (a) What is a drilling fluid ? What are the functions of drilling fluids ? (CO3)

OR

- (b) What is the purpose of casing used in oil and gas well drilling operations ? Describe the different types of casing. (CO3)

Roll No.

TMA-307

B. TECH. (PETROLEUM ENGINEERING)
(THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

ADVANCED MATHEMATICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Check the function $u(x, y) = 2x - x^3 + 3xy^2$ is harmonic function or not.

If yes find its harmonic conjugate. (CO2)

OR

- (b) Solve the given difference equation using Z-transform method : (CO1)

$$y_{n+2} - 3y_{n+1} + 2y_n = 0; \text{ given } y_0 = 0 \text{ and } y_1 = 1.$$

2. (a) Check the analyticity of the following functions : (CO2)

(i) $f(z) = |z|^2$ for all z .

(ii) $f(z) = z^2$ for all z .

P. T. O.

OR

- (b) Define the Z-transform. Find the Z-transform of a discrete unit step sequence : (CO1)

$$u(n) = \begin{cases} 1 & \text{for } n \geq 0 \\ 0 & \text{for } n < 0 \end{cases}$$

3. (a) Show that the function $f(z) = u + iv$, where

$$f(z) = \begin{cases} \frac{x^2 y^5 (x + iy)}{x^4 + y^{10}}, & z \neq 0 \\ 0, & z = 0 \end{cases} \quad \text{is not analytic at } z = 0 \text{ although}$$

Cauchy-Riemann equations are satisfied at $z = 0$. (CO2)

OR

- (b) Find the Z-transform of $\{f(n)\}$, where $f(n) = \begin{cases} 5^n & \text{for } n \geq 0 \\ 3^n & \text{for } n < 0 \end{cases}$.

(CO1)

4. (a) Using Partial fraction method to find the Z-transform of the given

$$\text{function : } F(z) = \frac{3z^2 - z}{(z-1)(z-2)^2} \quad (\text{CO1})$$

OR

- (b) Define analytic function. Show that the function $f(z) = z|z|$ is not analytic anywhere. (CO2)

(3)

5. (a) Find inverse Z-transform of the function $F(z) = \left(\frac{2z^2 + 3z}{(z+2)(z-4)} \right)$.

(CO1)

OR

- (b) Using Milne-Thomson's method determine an analytic function $f(z)$ where the real part of $f(z)$ is given $u(x,y) = \sin x \cosh y + 2 \cos x \sinh y + x^2 - y^2 + 4xy$.

(CO2)